PROJECT REPORT OF CO-OP Project at Industry (Module-2)
ON
SMART STUDY PLANNER

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING
In
COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"Smart Study Planner"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Shikha Tuteja**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

Date: ..**28-04-2026**..

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

The project titled **“Smart Study Planner – An AI-Powered Intelligent Learning and Scheduling System”** aims to design and develop a comprehensive full-stack web application that helps students plan, organize, and optimize their study schedules efficiently. The system automates study management by intelligently prioritizing subjects, generating adaptive schedules, tracking progress, and improving productivity through AI-driven recommendations.

The primary objective of the project is to eliminate inefficient manual study planning methods and provide students with a centralized platform for managing subjects, topics, study tasks, deadlines, and academic progress. The application analyzes exam timelines, topic dependencies, learning stages, complexity levels, workload distribution, and available study hours to create optimized and realistic daily study plans.

The system incorporates secure authentication mechanisms including JWT-based authentication, OTP verification, password recovery, and persistent login sessions to ensure user security and reliability. Students can create subjects with exam dates, add and organize topics, track completion status, and monitor workload metrics and progress analytics in real time.

One of the key features of the application is the AI-powered topic metadata generation system. The platform uses the Gemini API to generate structured metadata for topics, including importance, complexity, dependency levels, learning stages, and prerequisite relationships. If the AI service becomes unavailable, the system automatically falls back to a heuristic-based metadata generation algorithm to maintain uninterrupted functionality.

The application implements an advanced scheduling engine that dynamically distributes study time across subjects based on urgency, workload, remaining time before exams, and topic prioritization rules. The scheduling logic ensures efficient allocation of available study hours while identifying overloaded subjects and calculating realistic completion percentages

1. Introduction

Overview of the Project

Smart Study Planner – An AI-Powered Intelligent Learning and Scheduling System is a web-based application designed to modernize and simplify the study planning process for students. In many academic environments, students still manage their study schedules manually, which often leads to poor time management, uneven workload distribution, missed deadlines, and reduced productivity.

The proposed system provides a comprehensive digital platform that integrates subject management, intelligent scheduling, task tracking, and progress monitoring into a single unified application. By leveraging modern web technologies and AI-powered recommendation systems, the platform helps students organize their learning activities efficiently and generate optimized study schedules based on exam dates, workload, topic dependencies, and available study hours.

The system centralizes key functionalities such as subject creation, topic prioritization, automated timetable generation, and real-time progress tracking. This centralization improves study consistency, reduces manual planning effort, and enables students to focus more effectively on important topics. Additionally, the application is designed with a user-friendly and responsive interface, ensuring smooth interaction for users across different devices and technical skill levels.

Overall, Smart Study Planner contributes to the digital transformation of educational productivity by improving study efficiency, minimizing scheduling conflicts, and enhancing academic performance through intelligent automation and adaptive learning management.

Problem Statement

Despite the availability of digital learning tools, many students still rely on traditional or unorganized methods for planning and managing their studies. These methods often create problems such as poor time management, uneven subject preparation, missed deadlines, and difficulty in tracking academic progress. Manual study planning also makes it challenging for students to balance workloads across multiple subjects and exams effectively.

Existing study planner applications often provide only basic scheduling features and lack intelligent prioritization, dependency management, and adaptive workload distribution. Most systems fail to consider important factors such as exam urgency, topic complexity, learning stages, prerequisite relationships, and available study hours while generating study schedules. As a result, students may spend excessive time on less important topics while neglecting critical subjects.

Another major challenge is the absence of intelligent recommendation systems that can dynamically adjust study plans based on user progress and remaining workload. Many existing platforms also lack integrated progress tracking, task synchronization, gamification, and social motivation features, reducing user engagement and consistency.

Therefore, there is a strong need for a scalable, secure, and intelligent study planning system that can automate scheduling, optimize study time allocation, and provide personalized learning recommendations. The system should ensure efficient workload management, real-time progress tracking, adaptive schedule generation, and an interactive user experience that helps students improve productivity and academic performance.

Objectives of the Project

The primary objective of Smart Study Planner is to develop an intelligent and reliable study management system that simplifies and automates the process of study planning, scheduling, and progress tracking. The system is designed with the following specific objectives:

● To create a centralized platform for managing subjects, topics, schedules, tasks, and academic progress efficiently

● To minimize manual study planning efforts and reduce the chances of poor time management and uneven workload distribution

● To implement secure authentication mechanisms using modern technologies such as JWT, OTP verification, and password recovery to ensure user security and data protection

● To generate intelligent and adaptive study schedules based on exam deadlines, topic complexity, dependencies, and daily available study hours

● To provide AI-powered topic metadata generation and recommendation systems for better prioritization and learning management

● To enable efficient task tracking, topic completion monitoring, and real-time progress analysis for improved productivity

● To design a user-friendly and responsive interface that enhances usability and accessibility across different devices

● To include gamification and social features such as points, friend requests, and leaderboards to increase student motivation and engagement

● To ensure scalability of the system so that additional intelligent learning and productivity features can be integrated in the future

These objectives collectively aim to improve study efficiency, optimize workload management, and enhance students' academic performance through intelligent automation and adaptive scheduling.

Scope of the Project

The scope of Smart Study Planner includes the design and development of a fully functional AI-powered study management and scheduling system that addresses the core academic planning needs of students. The system primarily focuses on managing subjects, organizing topics, generating intelligent study schedules, tracking progress, and improving productivity through automated recommendations.

The application allows users to register, log in, manage their profiles, and configure their daily study availability. Students can create subjects with exam dates, add multiple topics, generate AI-based topic metadata, and receive personalized study recommendations based on complexity, dependencies, workload, and urgency.

The system also provides automated schedule generation, task management, progress tracking, points and leaderboard systems, and social interaction features such as friend requests and weekly rankings. These functionalities help students maintain consistency, monitor performance, and stay motivated throughout their learning process.

While the current system covers all essential functionalities related to intelligent study planning and scheduling, it is designed with scalability in mind. Future enhancements can include mobile application support, advanced AI-based learning analytics, voice assistants, calendar synchronization, collaborative group study features, and integration with educational platforms.

The project is currently implemented as a web-based platform; however, it can be extended to mobile devices to improve accessibility and user engagement. Additional integrations with external educational tools and cloud services can further enhance the platform’s capabilities.

In summary, the scope of this project is to provide a strong foundation for a modern intelligent study planning system that can evolve with technological advancements and the growing academic needs of students.

2. Methodology

2.1. Development Approach

The development of Smart Study Planner follows the Agile software development methodology, which emphasizes iterative progress, flexibility, and continuous improvement. Unlike traditional development models such as the Waterfall approach, Agile enables the project to evolve through multiple development cycles, allowing the team to adapt to changing requirements and improve features based on continuous testing and feedback.

The project was divided into smaller functional modules, and each module was designed, developed, tested, and refined through iterative development phases. Regular collaboration and discussions among team members helped identify issues early, improve scheduling logic, optimize AI integration, and enhance overall system performance incrementally. This approach ensured better coordination, faster debugging, and a more reliable final application.

Agile methodology also supported effective task distribution among team members, allowing parallel development of different components such as frontend UI design, backend APIs, database management, authentication systems, AI metadata generation, and intelligent scheduling logic. This collaborative development process improved productivity and ensured smooth integration of all project modules.

2.2 Requirement Analysis

Requirement analysis is a crucial phase in the development process, where the needs and expectations of users are carefully studied and documented. In this project, requirements were identified by analyzing the common challenges faced by students in managing study schedules, tracking progress, organizing subjects, and preparing for exams efficiently.

The requirements were categorized into two main types:

2.2.1 Functional Requirements

These define the core functionalities of the system, including:

- User registration, OTP verification, and secure authentication
- Subject and topic management with exam date handling
- AI-powered topic metadata generation and recommendation system
- Intelligent schedule generation based on workload and urgency
- Task management and progress tracking
- Points, leaderboard, and social interaction features
- Dashboard for viewing schedules, subjects, and performance metrics

2.2.2 Non-Functional Requirements

These focus on system performance and usability, including:

- Security and data protection using JWT authentication
- Scalability to support increasing users and future feature expansion
- User-friendly and responsive interface across devices
- Reliability and availability of scheduling and recommendation systems
- Efficient data handling and optimized performance

2.3 System Design

The system design phase involved creating the architecture and structure of the application to ensure efficiency, scalability, maintainability, and smooth integration of intelligent scheduling features

2.3.1 Architecture Design

The application follows a client-server architecture:

- The frontend (client-side) handles user interaction
- The frontend (client-side) handles user interaction, schedule visualization, and dashboard management
- The backend (server-side) processes authentication, business logic, AI metadata generation, scheduling algorithms, and API requests

2.3.2 Database Design

A structured database schema was designed to efficiently store data related to users, subjects, topics, schedules, tasks, friend requests, and leaderboard points. Proper relationships and constraints were defined to maintain data integrity, ensure synchronization between tasks and topics, and avoid redundancy

2.3.3 User Interface Design

The UI was designed to be modern, intuitive, and responsive, ensuring smooth navigation for students across different devices and screen sizes. The interface includes dashboards, subject management pages, schedule views, task tracking sections, and social features designed to improve usability and engagement. Wireframes and layouts were planned before implementation to maintain consistency throughout the application.

2.4 Development Process

The development phase involved implementing the designed system using modern web technologies and intelligent scheduling algorithms

2.4.1 Frontend Development

The frontend was developed using Next.js, which provides a component-based architecture for building dynamic, fast, and interactive user interfaces. Tailwind CSS was used for styling and creating a responsive design that works efficiently across different devices and screen sizes.

2.4.2 Backend Development

The backend was built using Node.js and Express.js, which handle server-side logic, API development, authentication, scheduling algorithms, and communication with the database. RESTful APIs were implemented to ensure smooth data exchange between the frontend and backend systems.

2.4.3 Database Integration

PostgreSQL was used as the primary database, and Prisma ORM was utilized for efficient database management, schema handling, and query execution. This enabled seamless interaction between the application and the database while maintaining data consistency and reliability

`2.4.4 Authentication and Authorization

JWT (JSON Web Tokens) was implemented to provide secure authentication and session management. Additional security features such as OTP verification, password recovery, and protected routes were integrated to ensure secure access to user data and application features.

2.5 Testing and Deployment

Testing is an essential phase to ensure the reliability, accuracy, and performance of the system. Various testing techniques were used to identify and resolve issues throughout the development process.

2.5.1 Testing

- **Unit Testing:** Individual modules such as authentication, scheduling logic, topic recommendations, task management, and API endpoints were tested to ensure correct functionality

- **Integration Testing:** Interaction between the frontend, backend, database, AI metadata generation system, and scheduling engine was verified to ensure smooth data flow and synchronization

- **User Testing:** The application was tested from a student's perspective to ensure usability, responsiveness, and ease of navigation across different devices. Bugs and errors identified during testing were fixed to improve system stability and performance.

2.5.2 Deployment

After successful testing, the application was deployed using cloud platforms:

- The frontend, built with Next.js, was deployed on Vercel, which provides optimized performance, automatic scaling, and seamless integration with the framework.
- The frontend, built with Next.js**,** was deployed on Vercel**,** which provides optimized performance, automatic scaling, and seamless integration with the framework
- PostgreSQL was used as the cloud-hosted database for reliable and scalable data storage and management

Deployment ensures that the application is accessible online and can handle real-world usage. Proper configuration and environment setup were done to maintain performance and security.

3 Tools and Technologies

3.1 Frontend

The frontend of Smart Study Planner is developed using Next.js and Tailwind CSS. Next.js provides a powerful React-based framework that supports server-side rendering, optimized routing, and high-performance web application development. It enables fast loading speeds, improved scalability, and efficient state management for dynamic study scheduling features.

Tailwind CSS is used for styling the user interface. It enables rapid UI development through utility-first classes, ensuring consistency, responsiveness, and a modern design across different screen sizes and devices. Together, these technologies help create an interactive, clean, and user-friendly learning platform.

3.2 Backend

The backend is implemented using Node.js and Express.js, which provide a scalable and efficient environment for handling server-side logic, scheduling algorithms, authentication, and RESTful API development. Express.js simplifies API creation and efficiently manages request-response communication between the frontend and backend.

The system also integrates the Gemini API for AI-powered topic metadata generation, including complexity analysis, dependency mapping, and learning-stage classification. Additionally, MongoDB services are used for authentication support, backend utilities, and scalable cloud-based operations.

3.3 Database

PostgreSQL is used as the primary relational database for storing structured data such as user information, subjects, topics, schedules, tasks, points, and social interactions. It ensures data integrity, reliability, and efficient query processing.

4 Implementation

4.1 Architecture

The Smart Study Planner system is designed using a client-server architecture, which ensures clear separation between the user interface and backend processing logic. This architecture improves scalability, maintainability, performance, and efficient handling of intelligent scheduling operations.

In this model, the frontend (client-side), built using Next.js, manages user interaction, dashboard rendering, schedule visualization, and responsive UI components. It communicates with the backend through secure RESTful APIs for seamless data exchange and real-time updates.

The backend, developed using Node.js and Express.js, processes API requests, authentication, business logic, AI metadata generation, recommendation algorithms, and intelligent schedule creation. It also manages task synchronization, progress tracking, and social features such as friend requests and leaderboards.

API communication plays a crucial role in the system, enabling efficient interaction between the frontend and backend. Operations such as user authentication, subject management, topic generation, task updates, and schedule regeneration are handled through secure API endpoints.

The system integrates the Gemini API for AI-powered topic metadata generation and fallback heuristic logic for uninterrupted scheduling functionality. PostgreSQL is used for structured data storage, while Prisma ORM and MongoDB services help manage authentication, database interaction, and cloud-based backend operations efficiently.

The overall architecture ensures that the system remains modular, secure, and easily extendable with future features such as mobile support, advanced analytics, and additional AI-based learning tools.

4.2 Features

The Smart Study Planner system incorporates a comprehensive set of intelligent features designed to automate study planning, optimize scheduling, improve productivity, and provide a secure and engaging learning experience. Secure JWT-based authentication with role-based access control, providing isolated permissions for admin, doctor, and staff roles to ensure data security and controlled system access.

- Secure JWT-based authentication system with OTP verification, password recovery, reset password functionality, and persistent login sessions to ensure secure user access and data protection.

- Intelligent subject and topic management system that allows users to create subjects with exam deadlines, add bulk topics, manage completion status, and automatically track academic progress.

- AI-powered topic metadata generation using the Gemini API, capable of generating structured topic information such as importance, complexity, dependency level, learning stage, and prerequisite relationships.

- Heuristic fallback metadata engine that automatically generates topic metadata and priority scores when AI services are unavailable, ensuring uninterrupted scheduling functionality.

- Advanced recommendation and prioritization engine that ranks topics using dependency-aware scoring logic, foundational topic detection, and workload analysis.

- Dynamic study scheduling system that automatically distributes available study hours across subjects based on exam urgency, remaining workload, and intelligent time allocation algorithms.

- Real-time task management and synchronization system where task completion automatically updates topic progress and subject completion status.

4 Results

5.1 Efficiency

The implementation of Smart Study Planner has significantly improved the efficiency of study management and academic planning. By automating tasks such as schedule generation, topic prioritization, workload distribution, and progress tracking, the system reduces the dependency on manual planning methods. This helps students save time, maintain consistency, and focus more effectively on important subjects and topics.

The intelligent scheduling engine ensures balanced study sessions based on exam urgency, workload, and available study hours. Automated recommendations and real-time task synchronization further improve productivity and reduce confusion during exam preparation.

5.2 Data Management

The system provides a centralized database for storing and managing all academic-related data, including user profiles, subjects, topics, schedules, tasks, points, and social interactions. This centralized approach eliminates redundancy and ensures consistency throughout the application.

Users can easily access, update, and track their study progress in real time. Structured data storage using PostgreSQL and Prisma ORM improves reliability, synchronization, and efficient retrieval of information whenever required. The integration of AI-generated metadata and scheduling information further enhances intelligent data handling within the system.

5.3 Security

Security is a critical aspect of the system, and Smart Study Planner ensures strong protection of user data through secure authentication and authorization mechanisms. JWT-based authentication, OTP verification, password recovery systems, and protected API routes ensure that only authorized users can access application features and personal study information.

Additional security measures such as encrypted password storage, secure session handling, and controlled API access help maintain user privacy and data integrity. Overall, the system provides a secure and reliable environment for managing sensitive academic and personal information.

5 Future Scope

The Smart Study Planner system provides a strong foundation for intelligent academic planning and productivity management; however, there are several opportunities for further enhancement and expansion to make the system more advanced and comprehensive.

● Integration with mobile applications and cross-platform synchronization for improved accessibility and real-time study management.

● Advanced AI-powered learning analytics to provide personalized study strategies, performance predictions, and adaptive learning recommendations.

● Integration with calendar platforms, reminders, and notification systems for automated deadline tracking and schedule alerts.

● Voice-assisted study management and AI chatbot support for interactive learning guidance and productivity assistance.

● Collaborative study features such as group scheduling, shared study rooms, and peer learning communities.

● Advanced visualization dashboards for detailed academic insights, workload forecasting, and long-term performance analysis.

● Integration with external educational platforms, online courses, and digital learning resources for centralized academic management.

● Enhanced gamification features including achievement systems, streak tracking, rewards, and personalized challenges to improve motivation and consistency.

● Offline support and cloud synchronization to ensure uninterrupted access to schedules and tasks across devices.

● AI-based exam preparation assistance with revision planning, smart quizzes, and automatic identification of weak topics.

These enhancements can transform Smart Study Planner into a fully intelligent, adaptive, and scalable educational productivity platform capable of supporting diverse learning environments and future technological advancements.

6 Conclusion

The Smart Study Planner – An AI-Powered Intelligent Learning and Scheduling System successfully demonstrates how modern digital technologies and intelligent algorithms can transform traditional study planning methods into an efficient, automated, and adaptive learning experience. The system effectively addresses the limitations of manual study management by providing a centralized, secure, and intelligent platform for organizing subjects, scheduling study sessions, tracking progress, and optimizing academic productivity.

Through the implementation of modern web technologies such as Next.js, Node.js, PostgreSQL, Prisma ORM, and Supabase, the project achieves high performance, scalability, reliability, and responsive user interaction. Key functionalities such as AI-powered topic metadata generation, intelligent scheduling, workload balancing, progress tracking, gamification, and social learning features significantly improve study efficiency and user engagement.

The system reduces manual planning effort, minimizes scheduling conflicts, and ensures better time management through automated prioritization and adaptive scheduling algorithms. Additionally, the integration of the Gemini API enhances the platform by enabling intelligent topic analysis, dependency mapping, and personalized study recommendations.

This project demonstrates the effective collaboration of the development team in designing, implementing, and integrating frontend, backend, database, AI, and scheduling modules into a cohesive full-stack application. The structured development approach and modular architecture played an important role in the successful completion of the system.

In conclusion, Smart Study Planner provides a practical, scalable, and intelligent solution for modern academic planning and serves as a strong foundation for future advancements in AI-powered educational productivity systems.

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- GitHub Actions Documentation. Available at: https://docs.github.com/en/actions
- React Documentation. Available at: https://react.dev

8 Appendices

Appendix A: System Screenshots

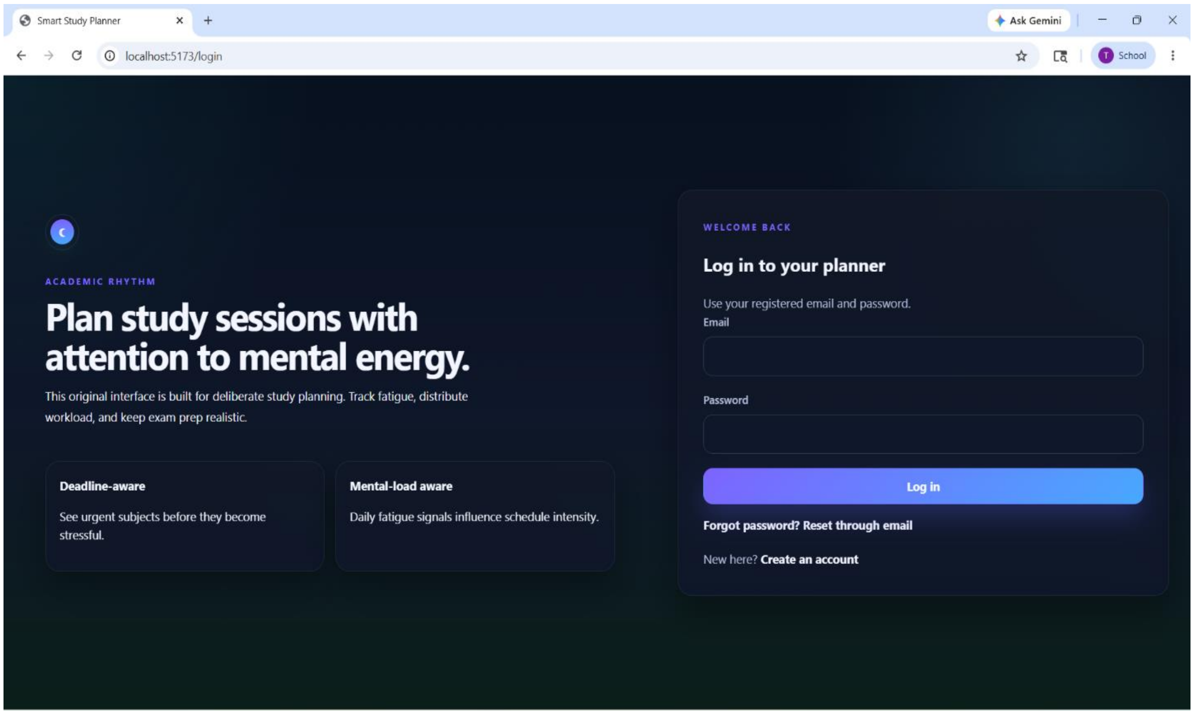

Figure A.1: Login Page

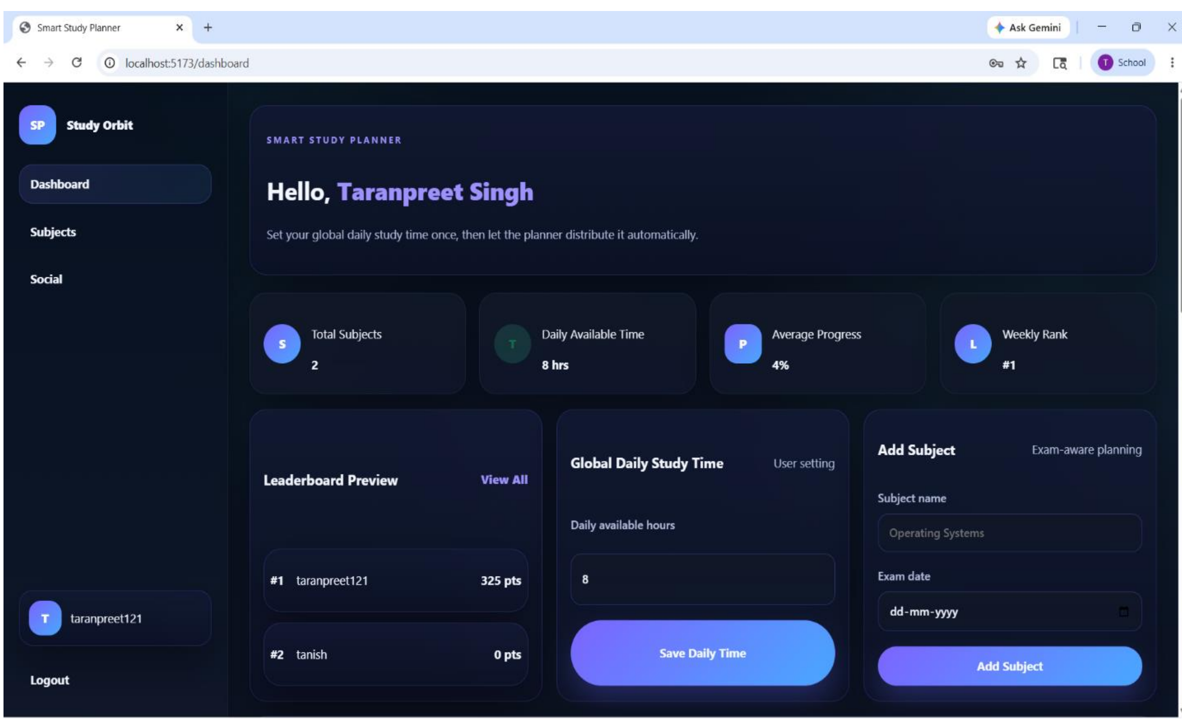

Figure A.2: Dashboard

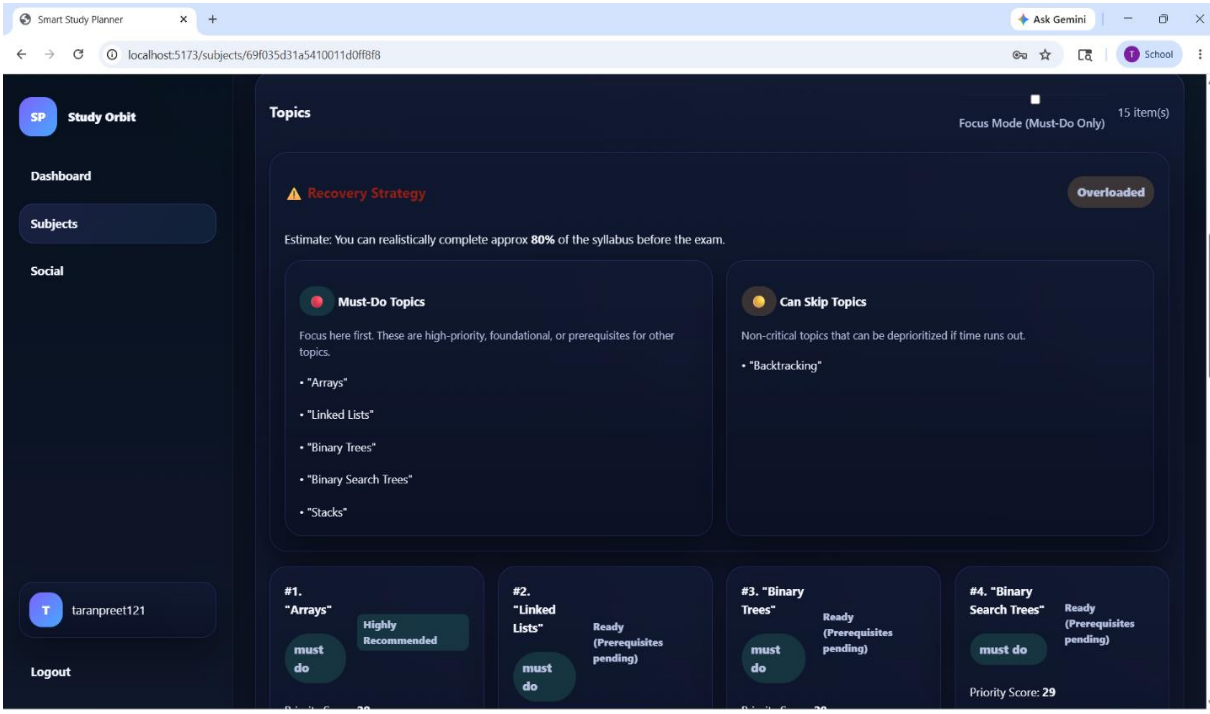

Figure A.3 Subjects Dashboard

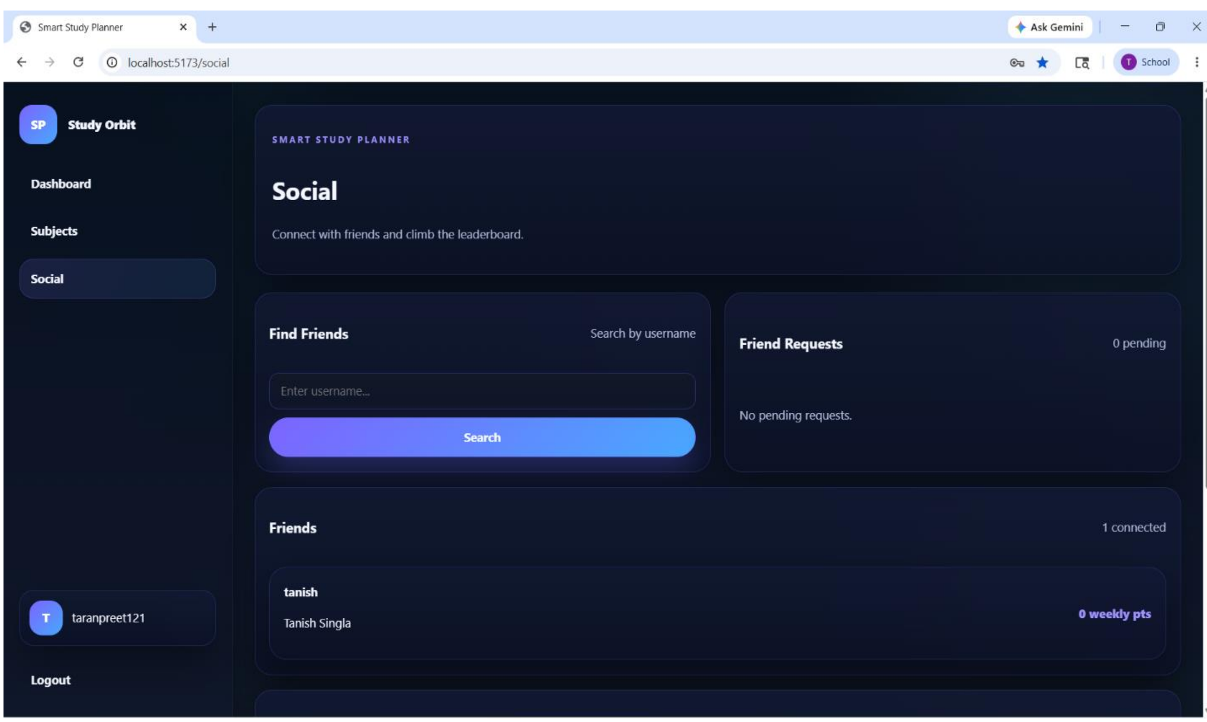

Figure A.4: Social Overview